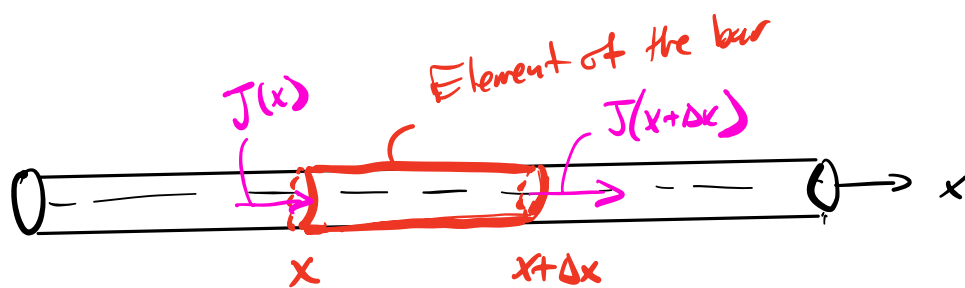
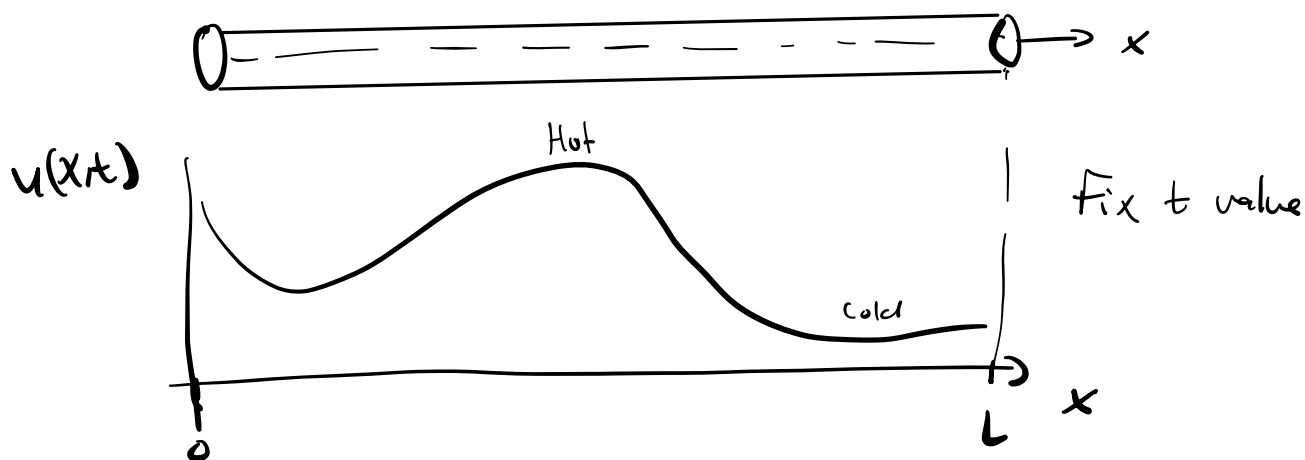


$u(x,t)$ = temperature at space value x , at time t



$$\sigma \rho A \Delta x u(x + \frac{\Delta x}{2}, t)$$

Annotations for the equation above:

- σ : specific heat (heat/mass · deg)
- ρ : mass density (mass/vol)
- $A \Delta x$: Volume
- $u(x + \frac{\Delta x}{2}, t)$: approx temp (degrees)

Change in heat in this element

$$\frac{\partial}{\partial t} (\sigma \rho A \Delta x u(x + \frac{\Delta x}{2}, t))$$

$$\Rightarrow \boxed{\sigma \rho A \Delta x u_t(x + \frac{\Delta x}{2}, t)}$$

Flux:

$$J(x) - J(x + \Delta x)$$

Fourier's law:

$$J \propto -\frac{\partial u}{\partial x}$$

$$J = -k u_x$$

$$-k u_x(x, t) + k u_x(x + \Delta x, t)$$

$$\sigma_p A \Delta x u_t(x + \frac{\Delta x}{2}, t) = -k u_x(x, t) + k u_x(x + \Delta x, t)$$

$$u_t(x + \frac{\Delta x}{2}, t) = \frac{k}{\sigma_p A} \left(\frac{u_x(x + \Delta x, t) - u_x(x, t)}{\Delta x} \right)$$

$$\lim_{\Delta x \rightarrow 0} u_t(x, t) = \frac{k}{\sigma_p A} u_{xx}(x, t)$$

— PDE

$$\text{Heat Eqn: } u_t = \alpha u_{xx}$$

~ IVPBP

$$u_t = \alpha u_{xx} \quad t \in (0, \infty), x \in (0, L)$$

$$\frac{\partial u}{\partial x} \Big|_{x=0} = 0 \quad t \in (0, \infty), x=0$$

$$\frac{\partial u}{\partial x} \Big|_{x=L} = 0 \quad t \in (0, \infty), x=L$$

$$u(x, 0) = f(x) \quad t=0, x \in [0, L]$$

Heat Eqn with Neumann BC's